

Table 1: Relationship between sensitivity, resolution, and uncertainty in inverse problems.

Concept	Mathematical basis	Interpretation
Sensitivity	$\mathbf{J} = \frac{\partial \mathbf{f}}{\partial \mathbf{m}}$	Describes how strongly predicted data respond to changes in model parameters
Uncertainty	$\mathbf{C}_m = (\mathbf{J}_w^T \mathbf{J}_w + \lambda \mathbf{L}^T \mathbf{L})^{-1}$	Quantifies confidence in model parameters and parameter correlations
Resolution	$\mathbf{R} = \mathbf{C}_m \mathbf{J}_w^T \mathbf{J}_w$	Describes how the true model is filtered or smeared in the recovered model